Entanglement of a 20-qubit GHZ state on MIMIQ

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We prepared $|\mathrm{GHZ}\rangle = (|0\cdots0\rangle + |1\cdots1\rangle)/\sqrt{2}$ on $N = 4$ to 20 qubits (Fig. 1). Cut measures: MIMIQ cloud, MPS simulator (the local MPS simulator had no free licence seat); everything else on the laptop. Clusters: $\rho = 2^{-k}\sum_P \langle P\rangle P$ from Pauli expectation values (Exaqt), entanglement from the thesis GCG code.

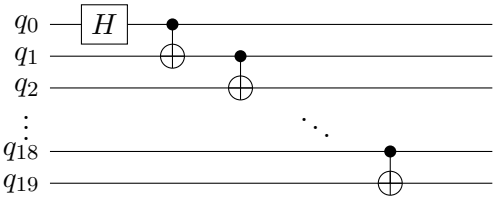

Figure 1: Circuit preparing the GHZ state on a chain of N qubits.

For a cut at qubit k ($A = \{0, \ldots, k-1\}$, B the rest), the two parts share one maximally entangled bit, $S = \log_2 r = 1$, for $N = 6, 8, 10$ and 20 alike (Table 1).

cut k	entropy $S(\rho_A)$ (bits)	Schmidt rank	bond dimension
0 (A empty)	0	1	1
1 to $N-1$	1	2	2

Table 1: Bipartite measures between A = qubits $0, \ldots, k-1$ and the rest of the chain (MIMIQ cloud, MPS); identical for $N = 6, 8, 10, 20$.

Every cluster is the classical mixture $(|0\cdots0\rangle\langle0\cdots0| + |1\cdots1\rangle\langle1\cdots1|)/2$: no entanglement, for neighbouring and for distant qubits (Table 2).

qubits	Pauli strings	purity	measure	value
0, 1	15	0.5	bipartite (0 vs 1)	0
0, 1, 2	63	0.5	$\mathcal{C}_{\mathrm{GME}}$	0
0, 1, 2, 3	255	0.5	$\mathcal{C}_{\mathrm{GME}}$	0
0, 10, 19	63	0.5	$\mathcal{C}_{\mathrm{GME}}$	0

Table 2: Entanglement inside clusters of the 20-qubit GHZ state, all other qubits ignored (traced out); Pauli expectation values (Exaqt), GCG convex roof.

For the whole state the GME concurrence is 1 at every size (Table 3). Up to $N = 8$ it is obtained from measurements only, with ρ rebuilt from all $4^N - 1$ Pauli strings, and it agrees with the value from the state vector. The number of strings grows as 4^N, so beyond $N \approx 8$ only the state-vector value is shown.

N	Pauli strings	$\mathcal{C}_{\mathrm{GME}}$ from measurements	$\mathcal{C}_{\mathrm{GME}}$ from state vector
4	255	1.000000	1.000000
6	4 095	1.000000	1.000000
8	65 535	1.000000	1.000000
10	1 048 575	–	1.000000
16	4.3×10^9	–	1.000000
18	6.9×10^{10}	–	1.000000

Table 3: GME concurrence of the full GHZ state (pure-state formula, all $2^{N-1} - 1$ bipartitions). State-vector values are from the simulator memory, not from measurements.

GHZ entanglement is fully collective: maximal across every cut, absent in every smaller group. For large N it cannot be measured through the nBRDMs; entanglement witnesses or spin inequalities are the practical route (to discuss with QPerfect).